1 Terrestrial Reference Frames, Epochs, and Datum Transformation

A terrestrial coordinate is three numbers, a frame, and an epoch. The engine works in ITRF at the epoch of observation, so the frame and the epoch are usually implicit. They become explicit when a result is compared with a national datum, a published datasheet, or a passive mark. Changing the frame and changing the epoch are different operations, and mixing them up produces decimetre-level offsets. The rule is:

$$\boxed{\textbf{Frames take a Helmert. Epochs take a velocity.}} \tag{1}$$

Sections 1.1 to 1.3 develop the rule and its one exception. Sections 1.4 and 1.5 give the sign conventions and the published parameter sets. Section 1.6 covers NATRF2022. Sections 1.7 and 1.8 cover the NGS Site Information File and the monument-to-ARP offset. Section 1.9 is a numerical example at CORV, and Section 1.10 compares the NGS deformation model with station velocities.

1.1 Frame versus epoch

A Helmert similarity (seven parameters, or fourteen with their linear rates) maps frame A to frame B *at the same epoch*. It describes how two realisations differ in origin, orientation, and scale. It says nothing about how the ground moves.

The fourteen-parameter form is

$$p(t) \;=\; p_0 \;+\; \dot{p}\,(t - t_0), \qquad p \in \{\boldsymbol{T},\, s,\, \boldsymbol{\varepsilon}\}, \tag{2}$$

and the rates $\dot{p}$ are rates *of the parameters*: the drift of one frame relative to the other. They are not station velocities. Evaluating (2) at epoch t gives the transformation from A to B at that epoch. It does not move the coordinate through time.

Moving a coordinate between epochs requires information about the site:

$$\boldsymbol{x}(t) \;=\; \boldsymbol{x}(t_0) \;+\; \boldsymbol{v}\,(t - t_0) \;+\; \boldsymbol{d}_{\mathrm{per}}(t) \;+\; \boldsymbol{d}_{\mathrm{PSD}}(t), \tag{3}$$

with $\boldsymbol{v}$ a station velocity, $\boldsymbol{d}_{\mathrm{per}}$ the periodic (annual + semiannual) terms, and $\boldsymbol{d}_{\mathrm{PSD}}$ the post-seismic deformation. `resolveStationCoordinate` and `tools/sif_propagate` implement this form. The three site terms are observed or modelled. A transformation parameter set does not contain them, so a transformation that appears to handle an epoch change is using the rigid-plate approximation of Section 1.3.

1.2 Transforming a velocity

Differentiating the linearised transformation with respect to time gives

$$\boldsymbol{v}_B \;=\; \boldsymbol{v}_A \;+\; \dot{\boldsymbol{T}} \;+\; \dot{s}\,\boldsymbol{x} \;+\; \dot{\boldsymbol{E}}\,\boldsymbol{x}. \tag{4}$$

Three properties of (4) matter in practice.

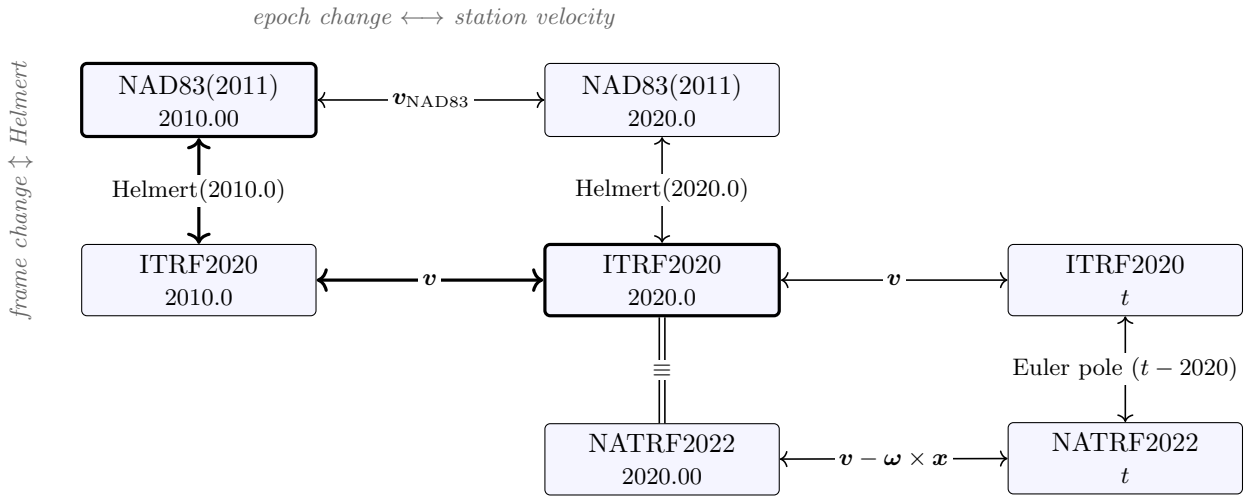


Figure 1: Frames (rows) and epochs (columns). Vertical moves change the frame at a fixed epoch; horizontal moves change the epoch within a frame and need a station velocity. Heavy lines mark the worked example of Section 1.9, from ITRF2020 at 2020.0 to NAD83(2011) at 2010.00. Because NAD83 rotates with the plate, $\boldsymbol{v}_{\mathrm{NAD83}}$ is small on the stable interior and larger at the plate margins (Section 1.3).

The rate field is rigid. The added terms $\dot{\boldsymbol{T}} + \dot{s}\boldsymbol{x} + \dot{\boldsymbol{E}}\boldsymbol{x}$ depend on the station's *position* only, not on its motion. This was checked by transforming fictitious stations at one position with different velocities: the added velocity was identical to the micron. The rates of a fourteen-parameter set are therefore a rigid velocity field, just as the seven parameters are a rigid displacement field.

Propagation and transformation commute, with one condition. Propagating in frame A and then transforming gives the same result as transforming and then propagating in frame B, to 0.000 mm, provided each propagation uses the velocity expressed in its own frame. That velocity comes from (4).

Using the wrong frame's velocity is a large error. Propagating a NAD83(2011) coordinate with an ITRF velocity costs about 20 mm/yr at CORV, or 203 mm per decade. No single-epoch check reveals it. The error exceeds every geophysical correction of the Site Displacement document combined.

1.3 The rigid-plate approximation

The parameter rates can stand in for a station velocity in one case. A plate-fixed frame is built so that stations on the rigid plate do not move in it. Setting $\boldsymbol{v}_B = \boldsymbol{0}$ in (4) gives

$$\boldsymbol{v}_A \;=\; -\big(\dot{\boldsymbol{T}} + \dot{s}\,\boldsymbol{x} + \dot{\boldsymbol{E}}\,\boldsymbol{x}\big). \tag{5}$$

The right-hand side is then a *modelled* rigid-plate velocity, and (2) is being used as a one-term deformation model.

The approximation is as good as the assumption. On stable North America the horizontal velocity relative to the plate is 1–2 mm/yr, so (5) costs 1–3 cm per decade. At a plate margin it fails. CORV sits on the deforming Cascadia margin, and the fourteen-parameter-only reconstruction of Section 1.9 misses by 104 mm per decade.

1.4 Rotation-sign conventions

Two rotation conventions are in use, and the sources cited here use both. The NGS / EPSG:1056 *coordinate frame* form, linearised for small rotations, is

$$\boldsymbol{x}' \;=\; \boldsymbol{x} \;+\; \boldsymbol{T} \;+\; s\,\boldsymbol{x} \;+\; \boldsymbol{E}\,\boldsymbol{x}, \qquad \boldsymbol{E} \;=\; \begin{bmatrix} 0 & \varepsilon_z & -\varepsilon_y \\ -\varepsilon_z & 0 & \varepsilon_x \\ \varepsilon_y & -\varepsilon_x & 0 \end{bmatrix}. \tag{6}$$

The IERS / IGN *position vector* form is identical except that $\boldsymbol{E}$ carries the opposite sign throughout; equivalently, $\boldsymbol{\varepsilon} \to -\boldsymbol{\varepsilon}$.

Using one convention's parameters in the other convention's matrix applies the rotation backwards. For a rotation angle θ and a position of magnitude L the error is $2|\theta|L$, twice the size of omitting the rotation. The same holds for the rate field of Section 1.2. A rate field derived here with the wrong convention came out exactly negated, a $\sim$35 mm/yr error at CORV where the true field is $\sim$ 20 mm/yr. Check the sign of a rate field against a station of known velocity; a plausible magnitude is not evidence of a correct sign.

1.5 Published parameter sets and their composition

Table 1 lists the three sets the tooling uses. The first is in position-vector convention and the other two are in coordinate-frame convention. The IGN rotations are zero, so the composition below can add the sets without converting between the conventions of Section 1.4.

Table 1: Published Helmert parameter sets. ITRF2020→ITRF2014 is IGN's, in position-vector convention; the two NAD83 sets are NGS/EPSG, in coordinate-frame convention. Rotations $\boldsymbol{\varepsilon}$ in milliarcseconds.

	ITRF2020→ITRF2014 IGN, position-vector $t_0 = 2015.0$	**ITRF2014→NAD83(2011)** NGS MYCS2, EPSG:8970 $t_0 = 2010.0$, coord-frame	**ITRF2020→NAD83(2011)** EPSG:10334 (pre-composed) $t_0 = 2010.0$, coord-frame
T_x	−1.4 mm	1.00530 m	1.00390 m
T_y	−0.9 mm	−1.90921 m	−1.90961 m
T_z	+1.4 mm	−0.54157 m	−0.54117 m
s	−0.42 ppb	0.36891 ppb	−0.05109 ppb
ε_x	0	26.78138 mas	26.78138 mas
ε_y	0	−0.42027 mas	−0.42027 mas
ε_z	0	10.93206 mas	10.93206 mas
$\dot{T}_x$	0.0 mm/yr	0.00079 m/yr	0.00079 m/yr
$\dot{T}_y$	−0.1 mm/yr	−0.00060 m/yr	−0.00070 m/yr
$\dot{T}_z$	+0.2 mm/yr	−0.00144 m/yr	−0.00124 m/yr
$\dot{s}$	0	−0.07201 ppb/yr	−0.07201 ppb/yr
$\dot{\varepsilon}_x$	0	0.06667 mas/yr	0.06667 mas/yr
$\dot{\varepsilon}_y$	0	−0.75744 mas/yr	−0.75744 mas/yr
$\dot{\varepsilon}_z$	0	−0.05133 mas/yr	−0.05133 mas/yr

PROJ applies the pre-composed EPSG:10334 set for `EPSG:9988`→`EPSG:6317`, with `t_epoch`= 2010 and the `coordinate_frame` convention. It can be rebuilt from the other two sets in two steps:

1. **Run the IGN set to the NGS epoch.** Evaluate (2) at 2010.0 from its own $t_0 = 2015.0$: $\boldsymbol{T}(2010) = \boldsymbol{T}(2015) + \dot{\boldsymbol{T}} \cdot (-5) = (-1.4,\, -0.4,\, +0.4)$ mm.

2. **Add term by term** to the NGS set at its native 2010.0.

The result matches PROJ's set: the largest difference over the fourteen parameters is 4×10^{-15} in the respective units. Adding the sets is valid because the cross terms of the exact composition are below a micron at Earth radius. The rotation-times-translation term $\boldsymbol{\Omega T}$ is $\sim$0.3 µm and the rotation-squared term $\Omega_1\Omega_2 R$ is $\sim$0.1 µm. The two-step chain and the pre-composed set agree to 0.0000 mm at 2010.0, 2015.0, and 2020.0.

Two conditions apply. First, both sets must be in the same rotation convention before they are added. That step is unnecessary here only because the IGN rotations are zero. Second, step 1 is required: adding the IGN set at its own 2015.0 reference epoch introduces a 0.5 mm error in Y and 1.0 mm in Z.

The EPSG codes are ITRF2020 `9988` / `9990` / `9989` (geocentric XYZ, 2D, 3D), ITRF2014 `7789` / `9000` / `7912`, and NAD83(2011) `6317` / `6318` / `6319`.

1.6 NATRF2022

NATRF2022, the North American frame of NSRS2022, is ITRF2020 rotated about the North America Euler pole. It has no translation and no scale. The two frames are identical at epoch 2020.00 and separate as the plate moves. With $\boldsymbol{\omega}$ the Euler vector, positive counter-clockwise under the right-hand rule,

$$\boldsymbol{x}_{\mathrm{NATRF}}(t) = \boldsymbol{x}_{\mathrm{ITRF2020}}(t) \; - \; (\boldsymbol{\omega} \times \boldsymbol{x})\,(t - 2020.0), \tag{7}$$

$$\boldsymbol{v}_{\mathrm{NATRF}} = \boldsymbol{v}_{\mathrm{ITRF2020}} - \boldsymbol{\omega} \times \boldsymbol{x}. \tag{8}$$

The NGS BETA EPP2022 Euler poles are, in mas/yr, $\boldsymbol{\omega}_{\mathrm{NA}} = (0.046, -0.704, -0.047)$, with the companion plates $\boldsymbol{\omega}_{\mathrm{PA}} = (-0.409, 1.063, -2.188)$, $\boldsymbol{\omega}_{\mathrm{CA}} = (-0.056, -0.957, 0.589)$, and $\boldsymbol{\omega}_{\mathrm{MA}} = (-8.089, 5.937, 2.159)$ for PATRF2022, CATRF2022, and MATRF2022 respectively. These are BETA values from `https://beta.ngs.noaa.gov/NATRF2022/`. Fetch the current values before production use.

The sign of (8) was checked on stable-interior stations, whose plate-fixed velocity must be near zero. The NATRF2022 velocities, ENU in mm/yr, are AMC4 (−0.49, 1.01, −1.17), GODN (−0.12, 0.47, −1.49), TXAU (−0.34, 1.73, −0.52), WES2 (0.07, −0.03, −0.12), and MIA3 (0.28, 0.90, 0.56). A sign error would double the plate rate, giving 20–30 mm/yr here.

PROJ 9.5.1 has no NATRF2022 CRS, so the caller must apply (7) directly.

1.7 SIF occupations and the datasheet convention

The NGS Site Information File (SIF) gives ITRF2020 coordinates at epoch 2020.0 from the MYCS3 solution. A station can have several **occupation** windows, each with its own reference coordinate and velocity. An occupation boundary is an equipment change or a known offset, so the coordinate series is discontinuous there. Two consequences follow.

The datasheet convention. NGS produces a datasheet's NAD83(2011) epoch-2010.00 coordinate as follows. Take the *latest* occupation. Propagate its reference coordinate back ten years with its linear velocity and **no** periodic term. Apply the fourteen-parameter transformation of Section 1.5 at 2010.0. This recipe reproduces the datasheet to 0.4 mm at CORV (PID AH2488) and 0.6 mm at CHZZ (PID AJ6961). In `tools/sif_propagate` it is the `--datasheet` option.

The engine convention. `resolveStationCoordinate`, and `tools/sif_propagate` by default, select the occupation that *covers the requested epoch*. A coordinate for an observation session must describe the equipment installed at that time. At CORV a request for 2010.0 uses the 1998–2019 occupation and differs from the datasheet by 32.8 mm. Of that, $\delta U = -32.4$ mm is the jump at the 2019-07-24 antenna change (ASH700936E_C → TRM59800.99 SCIS). Both coordinates are correct for their purpose. One rule follows:

Remark 1. The difference of two resolved coordinates across an occupation break is *not* a velocity. It includes the equipment offset, which is often the larger part.

Periodic terms are a constant offset at whole-year epochs. The SIF annual and semiannual terms are referenced to the station's reference epoch. A whole number of years from that epoch, cos = 1 and sin = 0, so the terms add a fixed vector: 0.65 mm at CORV, 1.73 mm at NEWP, 3.18 mm at CHZZ. The offset cancels in a difference between two such epochs but not in an absolute comparison. Published coordinates exclude it, so a propagated coordinate that includes it differs from the datasheet by these amounts.

1.8 Monument versus ARP

A datasheet comparison must also use the same reference point. SIF and datasheet coordinates refer to the **monument** (ground reference point, GRP). The engine's position state and the NGS CORS coordinate files refer to the **ARP**. The offset between them is the RINEX `ANTENNA:DELTA H/E/N` eccentricity, which the SIF publishes as `ARP_MINUS_GRP_ENU`. The reduction is developed in the marker/ARP section of the Coordinate Frames document.

At CORV, $\delta H = 1.521$ m, and both comparisons close:

- SIF monument + 1.521 m Up versus the published ITRF2020 ARP (−2498424.007, −3802820.797, 4454737.751): 0.3 mm.
- Datasheet + 1.521 m Up versus the published NAD83(2011) ARP (−2498423.096, −3802822.020, 4454737.741): 0.5 mm.

Omitting this step at CORV gives an obvious 1.5 m height error. The harder case is a site with an eccentricity of a few centimetres, where the error is small enough to go unnoticed.

1.9 Worked example: CORV

This example works every step above with numbers, at the monument, using the latest occupation (2019-08-04 to open). `tools/sif_propagate/corv_worked_example.py` reproduces it.

Starting point (SIF, ITRF2020 @ 2020.0). $\boldsymbol{x} = (-2498423.4121, -3802819.8917, 4454736.6830)$ m, $h = 105.5881$ m; velocity in ECEF (−9.8836, 2.3812, −3.3134) mm/yr, i.e. ENU (−9.57, −4.77, +0.12) mm/yr.

Step 2a: propagate to 2010.0 in ITRF2020. $\boldsymbol{x} - 10\boldsymbol{v} = (-2498423.3133, -3802819.9155, 4454736.7162)$ m; the station moved 106.9 mm.

Step 2b: ITRF2020 → ITRF2014 at 2010.0. With $\boldsymbol{T} = (-1.4, -0.4, +0.4)$ mm (the IGN set run to 2010.0) and $s = -0.42$ ppb: (−2498423.3136, −3802819.9143, 4454736.7147) m.

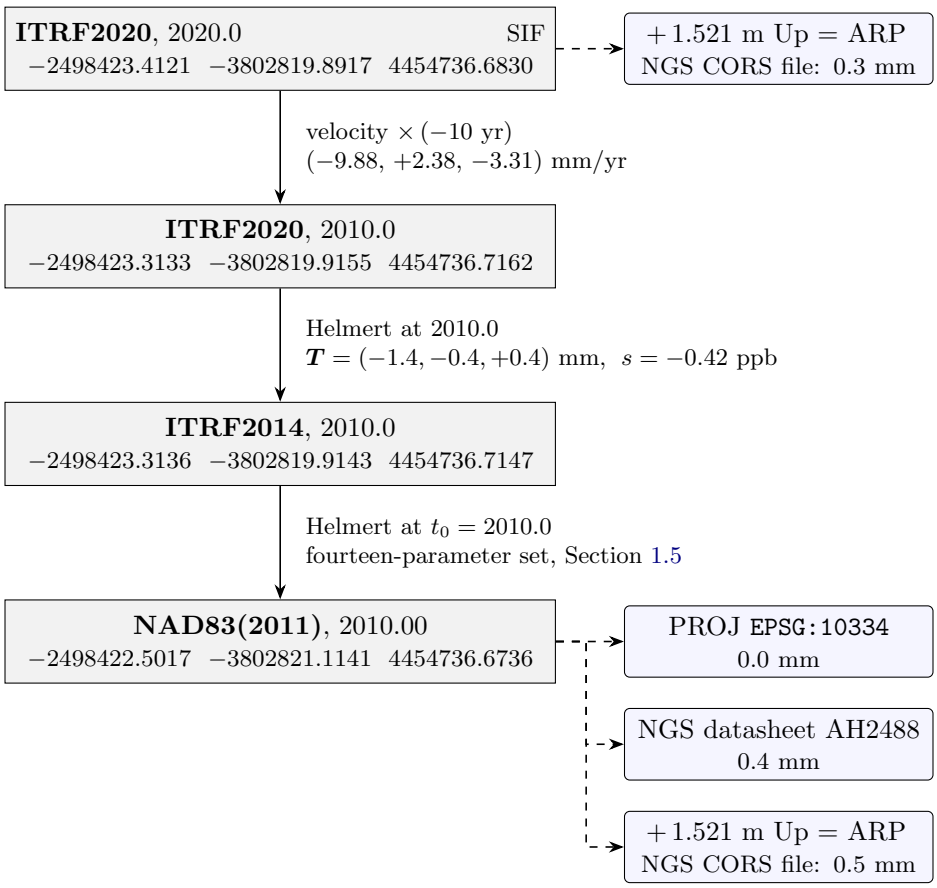


Figure 2: The CORV worked example, modern → legacy, at the monument (metres). Solid arrows are the computation; dashed arrows are independent checks, with the 3-D closure. The antenna reference point lies 1.521 m above the monument (RINEX `DELTA H`).

Step 2c: ITRF2014 → NAD83(2011) at 2010.0. Translation (1.00530, −1.90921, −0.54157) m; rotation term (−0.19247, 0.71082, 0.49885) m; scale term (−0.00092, −0.00140, 0.00164) m. Result (−2498422.5017, −3802821.1141, 4454736.6736) m. PROJ gives the same result (0.0 mm).

Closure against the datasheet. NGS datasheet AH2488 gives $\varphi = 44^\circ\,35'\,07.91300''$ N, $\lambda = 123^\circ\,18'\,16.51681''$ W, $h = 105.953$ m, i.e. (−2498422.5015, −3802821.1143, 4454736.6734) m. Hand computation minus datasheet is $\delta E = -0.3$, $\delta N = +0.2$, $\delta U = +0.1$ mm, which is 0.4 mm in three dimensions.

The reverse direction. Going from the datasheet to ITRF2020 at 2020.00, which is also NATRF2022 at that epoch (Section 1.6), requires ten years of station velocity. The miss against the SIF coordinate depends on the source of that velocity:

Table 2: CORV reverse reconstruction: miss against the SIF coordinate by velocity source. The last row is the rigid-plate approximation of Section 1.3, which does not hold at CORV.

Velocity source	**3D miss**	**Dominant component**
SIF velocity	0.4 mm	$\delta E = 0.3$
IFDM2022 grid (−8.68, −4.78, +0.04) mm/yr ENU	9.3 mm	$\delta E = 9.2$
BETA NCAT	9.6 mm	δE 6.9, δN − 5.4, δU 4.0
14-parameter only ($\boldsymbol{v}_{\mathrm{NAD83}} := \mathbf{0}$)	104.2 mm	δE − 54.6, δN − 88.2, δU 10.4

The same point in four frames. CORV velocity, ENU mm/yr: ITRF2020 (−9.57, −4.77, 0.12); plate motion $\boldsymbol{\omega} \times \boldsymbol{x}$ (−13.20, −13.12, −0.04); NATRF2022 (3.63, 8.35, 0.17); NAD83(2011) (5.49, 8.80, −1.05). The NATRF2022 velocity is what remains after removing rigid North America. At a deforming margin it is not small: a NATRF2022 coordinate at CORV moves (21.8, 50.1, 1.0) mm by 2026.0 and (36.3, 83.5, 1.7) mm by 2030.0 from its 2020.00 value. A plate-fixed coordinate is not a static one.

1.10 Deformation models and validation

A passive mark has no observed velocity, so a model must supply one. NGS supplies IFDM2022 with BETA NCAT. It is a global 0.1° ENU velocity grid with per-node standard deviations, in ITRF2020 (`epsg:9990`), plus 33 earthquake displacement grids. It was compared with SIF velocities at 21 stations in CONUS, Alaska, and British Columbia, using one occupation per station:

- $\|\boldsymbol{v}_{\mathrm{IFDM}} - \boldsymbol{v}_{\mathrm{SIF}}\|$: median 1.34 mm/yr, maximum 3.11 mm/yr.
- The grid's formal σ is optimistic by roughly a factor of two (median ratio 2.1, up to 5.9 in the stable interior). The comparison is *in-sample*, because the grid was built from these CORS. A passive mark should expect a larger error.
- In the stable interior the IFDM horizontal velocity equals the EPP2022 plate motion to 0.01 mm/yr. The model assigns zero intraplate velocity there.
- Velocity spread across a single grid cell is 0.03–0.05 mm/yr in the interior and up to 3.7 mm/yr in the Aleutians. Interpolation error matters only where the field is steep.

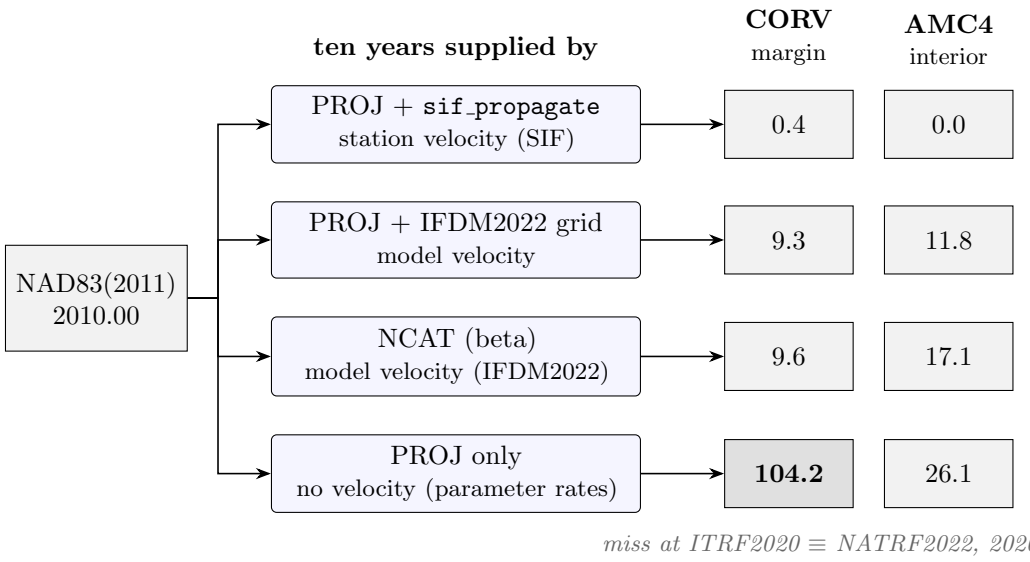


Figure 3: Legacy → modern, NAD83(2011) epoch 2010.00 to ITRF2020 ≡ NATRF2022 epoch 2020.00, by four routes. PROJ is the inverse of `EPSG:10334`, evaluated at 2010.0 (at 2020.0 in the last row, where it stands alone). The number is the 3-D miss against the SIF coordinate at 2020.0. CORV starts from NGS datasheet AH2488; AMC4 (Colorado Springs, single occupation) starts from its SIF-derived 2010.00 coordinate, so its first row is zero by construction.

NCAT's transformation from NAD83(2011) at 2010.00 to NATRF2022 at 2020.00 was compared with the SIF coordinate at 2020.0. The mean difference is (+4.9, −6.1, −1.3) mm and the median 3D difference is 18 mm, within NCAT's stated uncertainty of about 2 cm for most stations. The small mean confirms that NATRF2022 and ITRF2020 coincide at 2020.00 to a few millimetres, as Section 1.6 requires. Five stations are outliers: FAIR 115, SNI1 82, AB02 78, KYW1 61, and HOLP 43 mm. Four of them lie inside IFDM earthquake displacement grids. The SIF represents those events as occupation *breaks* rather than PSD terms, so the two sources model them differently. This attribution rests on grid coverage only. The displacement magnitudes were not evaluated, and KYW1 is unexplained.

Limitations of beta NCAT. It offers only fixed epochs: NAD83(2011) at 2010.00 and NATRF2022 at 2020.00. Its offline CLI performs coordinate *conversions* only, not datum transformations. Its XYZ input is rounded to 1 mm. The post-seismic terms of both tools are untested here, because no station in the comparison has a PSD term.

Mixing velocity sources. Within NSRS2022 the deformation model is the convention, so using IFDM (or HTDP) for a passive mark is correct by definition. The risk is mixing sources. A network that propagates its CORS with SIF velocities and its marks with grid velocities introduces a 1–2 mm/yr difference between the two groups, which distorts the network over a decade. The grid also gives only a diagonal covariance, and an optimistic one by the ratio above. SIF propagation gives the full

$$\boldsymbol{C}_{xx}(t) \;=\; \boldsymbol{C}_{xx} \;+\; \Delta t\,\big(\boldsymbol{C}_{xv} + \boldsymbol{C}_{xv}^{\mathrm{T}}\big) \;+\; \Delta t^{2}\,\boldsymbol{C}_{vv}, \tag{9}$$

including the position–velocity cross-covariance, which dominates (9) at short Δt and which the grid does not provide.